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I 100 PLANT CONTROL SYSTEM

1. SCOPE OF SUPPLY

The purpose of the technical specification is to define the design and functional requirements of control and monitoring system which fall within the scope of work as defined in Part 4, Subsection 1 .2.

It is the intent of the specification that fully functional control and monitoring system shall be supplied by the Contractor.

The Contractor shall supply the following instrumentation and control system with all components and complete accessories.

1.1 HUMAN MACHINE INTERFACE (HMI) EQUIPMENTS LOCATED IN CENTRAL CONTROL ROOM

For block 2 Central Control Room

- New Operator station in the central control room
- New HMI in the central control room for combine cycle
- New LCD 21" Monitor and PC base operator station in the central control room
- Emergency push button for steam Turbine trip, one button for each ST
- New 2 (two) color hard copier
- New 2 (two) Report Laser printer
- New 1 (one) printer Dot Matrix for SOE alarm printer
- The mimic diagram shall be extended to likely same with the Block 1 mimic diagram
- New HMI in the central control room for combined cycle

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For new block 3 & 4 Central Control Room

- 3 (three) 21" LCD monitor and PC based Operator Stations for each block
- 70" large screen VDU operation system for each block (3 & 4)
- GPS and clock system
- Emergency Push Button for Gas Turbine trip, one button for each GT
- Emergency Push Button for Steam Turbine trip, one button for each ST
- 1 (one) Main fire alarm panel (fire protection panel)
- 1 (one) Gas Detection Panel for each block
- 2 (two) Color Hard Copier for each block
- 2 (two) Report Laser Printer for each block

ADD1
-IIF
-002

- 1 (one) Dot Matrix Alarm Printer for each block
- 1 (one) CCTV Panel
- 1 (one) Clock Panel

1.2 ENGINEERING AND DIAGNOSTIC SYSTEM IN TECHNICAL STAFF ROOM

- 1 (one) Engineering station VDU
- 1 (one) Diagnosis station with VDU
- 1 (one) Event/Log laser printer

1.3 PLANT MANAGEMENT STATION

1.3.1 IN CHIEF OPERATOR ROOM

- 1 (one) Plant Management Station with VDU
- 1 (one) Color hard copier

1.3.2 IN PLANT MANAGER ROOM

- 1 (one) Plant Management Station with VDU
- 1 (one) Color hard copier

1.4 ST INSTRUMENTATION AND CONTROL SYSTEM LOCATED IN ST LOCAL CONTROL PACKAGES

For Steam Turbine

- 1 (one) Steam Turbine Supervisory Instrument (TSI) Panel
- 1 (one) Unit Protection Panel (Including HRSG Protection Relays)
- 1 (one) Steam Turbine Generator Protection Relay Panel
- 1 (one) Steam Turbine Generator Transducer Panel
- 1 (one) Steam Turbine Generator AVR Panel
- 1 (one) Generator synchronization facility for ST generator.

ST Local Control Packages should be located near the Steam Turbine area.

1.5 CONTROL SYSTEMS EQUIPMENTS LOCATED IN ELECTRONIC EQUIPMENT ROOM

1.5.1 DISTRIBUTED CONTROL SYSTEM (DCS) FOR THE FOLLOWING SYSTEMS

- Plant Co-ordination System
- HRSG Control System
- Electrical Control System
- BOP Control System
- Interface for RIO system

1.5.2 DATA LOGGING SYSTEM FOR EVENT RECORDER

1.5.3 HRSG TRANSDUCER PANEL

1.5.4 GATE WAY CONTROLLERS

- To Vibration Monitoring System and Continuous Emissions Monitoring System

1.6 REMOTE I/O SYSTEM FOR DCS

1.7 DATA COMMUNICATION SYSTEM FOR PLANT BUS, FIELD BUS, LAN AND COMMUNICATION BUS FOR GATE WAY.

1.8 DESKS FOR CONTROL EQUIPMENT LOCATED IN CCR & LOCAL CONTROL PACKAGES

- For Operator Station
- For Chief Supervisor
- For Engineering and Diagnostic System
- For ST Local Operator station

Layout of the Central Control Room shall be submitted by bidder for owner`s approval

1.9 THE FURNITURE, SUCH AS MEETING DESKS, CHAIRS, STORAGE CABINETS ETC.

Shall be included in Contractor`s scope of supply including but not limited to:

-	Operator Chairs	3 sets
-	Chief Supervisor Chair	1 set
-	Plant Manager desk and Chair	1 set
-	Engineer and Diagnostic Desks and Chairs	2 sets
-	Plant management Station Desk and Chairs	1 set
-	Meeting desk and Chairs	1 set
-	Shelves for the reference documents	1 set
-	Necessary Desks for Printers Hard Copys; etc	
-	Desk and Chair for CCTV (if any)	1 set
-	Necessary partitions to segregate the meeting Desk or engineers Desk.	1 set

2. **CODES AND STANDARDS**

The Contractor shall comply with the requirements of Part 4, subsection 2.5 “Codes and Standards”.

3. **DETAILED REQUIREMENT**

3.1 **GENERAL**

This specification describes the overall plant including operation philosophy, control and management concept.

Any works and equipment required for proper functioning of the system shall be included in the scope of works.

The control system for block 2 shall be separated from the former control system. The new dedicated DCS shall be provided for block 2 control system to avoid the complicated interfacing and apply legacy system. For block 3 & 4 control system, the existing GT control shall be integrated by DCS in the new Central Control Room.

The Contractor shall comply with this specification requirement.

The overall instrumentation and control system philosophy shall be based on the followings:

- a. High security for personnel and technological equipment

- b. Safe, reliable and efficient operation of the plant
- c. The equipment shall be able to be operated automatically from the Central Control Room (CCR) under all operation condition such as plant startup and abnormal condition and shall also apply to remote manual operation, normal operation and plant shutdown.
- d. The malfunction of any component or loss of control power supply shall lead to fail-safe operation.
- e. Redundancy and backup control shall be provided. The failure of a single component shall not cause an abnormal condition of the plant.
- f. The Instrumentation and Control (I&C) system for the Gas Turbine, Steam Turbine, HRSG, Electrical and BOP shall be integrated by the DCS. PLC model applied for the BOP and electrical control shall be provided from limited manufacturers.
- g. An historical data management system shall be provided for the management of historical data over along period.

3.2 PLANT OPERATION PHILOSOPHY

3.2.1 OPERATION CONCEPT

The combined cycle plant operation shall be based on full Visual Display Units (VDU) operation from Operator Stations.

A full automatic operation system shall be provided for the plant covering from the condition of “startup preparation completed” to normal operation, and from normal operation to plant shutdown. The condition of “startup preparation completed” is with maintenance work completed, no trip conditions, electrical and pneumatic feed available and water circuits full and ready.

A load run-back function shall be provided for reliable operation of the plant when predetermined fault conditions occur such as one of cooling water pump trip etc.

A “trip to house load” operation (GT operation without network connection) function shall be provided with rapid GT load reducing capabilities.

A GT simple cycle operation (GT operation without ST and/or HRSG

operation) function shall be provided with diverter damper “GT exhaust gas to bypass stack” operation.

A synchronization facility shall be provided to allow manual and automatic synchronization of GT generator circuit breaker and Steam Turbine generator circuit breaker.

The hardwired synchronizing equipment shall consist of a synchroscope, frequency and voltage indicators for incoming and running voltage and a synchronizing switch. This equipment shall be located in the Central Control Room.

For plant safety shutdown operation, the gas turbine, steam turbine and HRSG shall have manual emergency trip PB mounted on the associated Operator Station in the Central Control Room (CCR).

3.2.2 AUTOMATIC OPERATION

Plant full automatic operation shall be provided with at least the following four phases:

(1) Plant startup phase

From preparation of GT Start, such as maintenance work completed, no trip conditions, electrical and pneumatic feed available and water circuits full and ready, to target/rated load.

(2) Plant normal operation phase

Normal operation by Automatic Power Regulator (APR) or Load Frequency Control (LFC) demand from Load Dispatching Center (LDC).

(3) Plant shutdown phase

Plant normal operation to plant shutdown such as GT/ST turning gear operation.

(4) Plant re-starting phase

Plant re-starting and recovering operation after emergency condition such as House Load operation.

3.2.3 MANUAL OPERATION

All the closed loop control systems and open loop control systems of the DCS, such as Unit Co-ordination system, HRSG control system, BOP control system, and electrical control system, shall be able to be operated in manual from any operator station located in the Central Control Room (CCR).

Most of the closed loop controls for the GT control system shall not be able to be operated in manual or accept manual set points. Governor control and load limit control will always operate in auto, but will accept manual setpoints.

The remote operation shall be available from VDU Operator Station.

3.3 PLANT CONTROL PHILOSOPHY

3.3.1 CONTROL SYSTEM CRITERIA

For Control System Criteria, see PART 4 Technical Specifications I-005 “Distributed Control System”

3.3.2 HIERARCHICAL CONTROL SYSTEM CONCEPT

For Hierarchical Control System concept see PART 4 Technical Specifications I-001 “Instrumentation and Control”.

3.3.3 OPERATING AND SUPERVISING LOCATION

The unit shall be controlled and supervised from a number of operating and supervising control rooms as below:

3.3.3.1 CENTRAL CONTROL ROOM (CCR)

CCR shall be located as shown on the power station plot plan.

Control and supervision of the following equipment shall be carried out from the CCR:

- a) Gas Turbine and Gas Turbine Generator
- b) Steam Turbine and Steam Turbine Generator
- c) Heat Recovery Steam Generator
- d) BOP systems

- e) Plant Electrical Equipment
- f) Fire Protection System

3.3.3.2 SUBSTATION CONTROL ROOM

The Substation Control system shall be as specified in Part 4 Technical Specification E-460 “Substation Control System”.

The Sub-station Control System shall be connected to the Power Station Control System on the data highway.

3.3.3.3 BALANCE OF PLANT CONTROL SYSTEMS

Balance of Plant items are highly preferred controlled by DCS, but if controlled by specific control system, it shall also could be controlled and monitored by DCS from CCR.

3.3.4 ANNUNCIATION SYSTEM

The annunciation system for the process plant shall be integrated into a part of DCS system. See PART 4 Technical Specifications I-005 “Distributed Control System”

Fire alarm and security alarms shall be acquired from the fire fighting system.

3.3.5 SYSTEMS TO BE CONTROLLED

The DCS shall control some plant systems directly, and shall supervise the operation of others (including gas turbine control and steam turbine control systems). Each system shall provide the signal conditioning, data transfer, monitoring, closed loop control, open loop control, protection, operation and supervision as set out in PART 4 Section 3 Technical Specifications I-005 “Distributed Control System”. The controls shall include, but not be limited to, the following equipment and systems:

3.3.5.1 PLANT CO-ORDINATION CONTROL

- Automatic Plant Start up/Shut down control (APS)
- Automatic Power Regulation control (APR)

3.3.5.2 STEAM TURBINE CONTROL

Standard ST Control functions including governing

3.3.5.3 BOP CONTROL

The following BoP controls shall be supplied but not limited to:

- Gland steam pressure control.
- Gland steam temperature control
- Condenser hotwell level control
- Deaerator level control
- Deaerator pressure control

3.3.5.4 BOP AUXILIARIES CONTROL

Balance of Plant (BoP) items will be controlled by their own local control systems in the form of extensions of the DCS or of Programmable Logic Controllers (PLC). In the case of PLCs, they shall be connected to the DCS via data links and gateways.

Plant items are expected to include, but not be limited to, the following:

- Water Treatment System
- Raw Water System
- Waste Water System
- Dosing Systems
- Instrument Air System

3.3.5.5 HRSG CONTROL

The following HRSG controls shall be supplied but not limited to:

- High pressure steam turbine bypass control
- Low pressure steam turbine bypass control
- High pressure drum level control
- Low pressure drum level control
- High pressure superheater steam temperature control

- Reheater steam temperature control (if needed)
- Economizer inlet temperature control
- HRSG auxiliaries control

3.3.5.6 GENERATOR CONTROL SYSTEM

The following generator controls shall be supplied but not limited to:

- Automatic Voltage Regulator (AVR)
- Synchronising System

These are specified in Part 4 Section 3 E-101 It is expected that these controls will be “stand-alone” items connected to the DCS via the plant bus. In addition, numerical protection relays and energy meters will be connected to the plant bus so that the DCS can acquire their data (but not carry out their protection functions).

3.3.5.7 ELECTRICAL (POWER DISTRIBUTION) CONTROL

The following electrical controls shall be supplied but not limited to:

- 6.3 KV unit switchgear
- 380V unit switchgear
- 380V essential switchgear

In addition, numerical protection relays and energy meters will be connected to the plant bus so that the DCS can acquire their data (but not carry out their protection functions).

3.3.5.8 SYSTEMS TO BE INTERFACED

The DCS shall be interfaced with the following PLC (if any) but not be limited to:

- Water Treatment Plant
- Service Water and Potable Water System
- Waste Water Treatment Plant
- Chemical Dosing System
- HSD treatment Plant

- 500 kV Substation Control System

4. **PLANT CO-ORDINATION CONTROL SYSTEM**

The start up, operation within normal load range and shut down of the combined cycle plant shall be fully coordinated at the plant co-ordination control level. At this level the control functions in the control hierarchy shall be operated in unison to achieve the overall desired response.

For plant start up, the co-ordination level shall initiate the start up sequences of the major auxiliary systems in the appropriate order and intervals to minimize the run up period whilst maintaining the thermal stresses within given margins and maximizing the efficiency of the start up cycle.

Within the operational load range the co-ordination control level shall regulate the loading of the generators in accordance with the plant load set point control and the demand for frequency regulation as well as generating the sliding pressure set point for the HRSG steam outlet pressure.

4.1 **OPERATIONAL MODES**

The combined cycle plant shall be capable of operating in the following modes:

- (i) Independent operation of the gas turbine generator units exhausting to their bypass stack (i.e. open cycle operation)
- (ii) Gas turbine generator unit operating in the normal combined cycle mode

The transition from one operating modes into others operating mode shall be executed by operator initiation or automatically under fault conditions, to maintain maximum output and safety reason.

The control system shall provide, at the plant co-ordination level, all necessary controls, monitoring functions and sequential initiations to the sub-ordinate control levels to ensure the smooth transition between the states which shall be possible under all normal conditions.

It shall be possible for example, to transfer from open cycle to combined cycle operation without first reducing load on the gas turbine for hot start.

Any major constraints, which do not allow these requirements to be implemented, shall be identified by the Bidder.

4.1.1 CONTROL UNDER ABNORMAL CONDITION

4.1.1.1 GRID ISOLATED OPERATION

Should the plant be disconnected from the grid, the plant shall revert to isochronous control supplying the necessary power for the plant auxiliaries. Details of the method of control to achieve this method of operation shall be supplied by the Contractor. The plant shall have the ability for a quick restart from this condition. Black start capability is not required.

4.1.1.2 BYPASS OPERATION USING BY THE BYPASS DUMPER

If the ST or HRSG is out of service, the GT shall be able to run in bypass stack operation, by opening the diverter damper.

4.1.1.3 BYPASS OPERATION USING THE TURBINE BYPASS VALVE

If the ST is out of service, the GT and HRSG can run simultaneously. Then the generated steam is dumped into the condenser through turbine bypass valves. Due to the auxiliary power consumption, this operation mode shall only be used as an emergency procedure.

4.2 STEAM PRESSURE SET POINT CONTROL

Steam pressure control shall be done by two direct means:

- (i) Turbine Pressure Governing - this is the normal mode of control in which the turbine operates in the boiler following mode and the HP manifold pressure controller acts either directly on the turbine governor valves or indirectly via the governor speed set point.
- (ii) By-pass Pressure Governing - in this control mode the HP and LP manifold pressure controller act directly on the HP and the LP by-pass valves respectively.

In either mode of control the pressure set point and the transition from one mode to the other shall be functions of the plant co-ordination control level.

A sliding pressure characteristic is specified. This shall exhibit a constant pressure for start up and low load operation and then linear pressure increase with increasing steam flow until the maximum operating pressure is reached at MCR loading.

All necessary controls shall be provided for the convenient on load transition from the constant pressure characteristic to the variable pressure characteristic and in the reverse direction.

4.3 COORDINATED PLANT LOAD CONTROL

A setpoint for Plant MW load demand shall be provided at the plant co-ordination level. This shall regulate the power generated by the gas turbine generator units compensating appropriately for the power generated by the associated steam turbine. It shall be subject to the same load/frequency droop characteristic as the gas turbine generator. Plant Co-ordinated load control shall be operated with one, two or three gas turbine generators on combined cycle operation.

4.4 REQUIREMENT FOR LFC

The Muara Tawar Combined Cycle Power Plant may be used for Load Frequency Control (LFC) from Owner's Java Control Center.

The Contractor shall make provisions in his equipment to receive signals from the Owner's control centre. In the initial stage the Owner may operate the unit by providing the operator the load demand and load rate signals. These signals shall be used for digital display on the Operator Station. The operator will use this information to set the load demand and rate manually. Later these signals shall be utilized to set the load demand and rate automatically.

The Contractor shall be responsible to provide all the necessary interface hardware and software for this purpose. The following information shall be verified during detail design stage.

The load without the load frequency control is given by:

$$P = P_o + k \cdot df$$

where

$$P = \text{load}$$

Po = Load set point

1/k = Speed droop

df = Frequency deviation

The load with the load frequency control is given by:

$$P = P_o \pm N \cdot P_r \pm k \cdot df$$

where

Po = Load set point

Pr = Load control range of the unit

N = LFC level command order – $1 < N < +1$

The Contractor shall receive N, Po and Pr signals as 4~20 mA DC signals through the Owner's supervisory control and data acquisition (SCADA) system to be furnished by another Contractor. The signal ranges shall be verified during the detailed design. The Contractor shall receive these signals in his interface termination cabinet. The cabling and termination will be done by Contractor.

The Contractor shall provide two outputs from his control and monitoring system for interface with Owner's SCADA system in the form of electrically isolated (dry) normally open contacts which shall close for the conditions listed below:

- Unit trip
- Unit under constrained mode

These contacts shall be wired to terminal blocks in his interface termination cabinet located in substation control room. The SCADA system's remote terminal unit (RTU) will be connected to this interface cabinet.

The Contractor shall provide 4~20 mA DC signals representing unit generation (MW), unit master high limit (MW), unit master low limit (MW) and unit master rate limit (MW/mm) for other's use. These signals shall be terminated in the owner's interface termination cabinets. The signal range will be verified during the detail design.

The Contractor shall be responsible for any other signal required for the SCADA system. The detail of these signals will be given after Contract Award.

4.5 LOAD RUN-BACK CONTROL

Facilities shall be provided at the plant co-ordination control level to reduce the load demand to an appropriate level in the event of a major component failure such as a trip of gas turbine generator or HRSG.

4.6 SEQUENCE CONTROL FUNCTIONS

The following are the principal sequence control functions which shall be initiated through the plant co-ordination level, by single push button initiation.

4.6.1 COMBINED CYCLE PLANT START UP AND SHUT DOWN

The start up of the combined cycle plant shall be automated from the stage at which hot exhaust gas enters the Heat Recovery Steam Generator and steam is admitted to the steam turbine. Prior to start-up, a pattern of running plant shall be established by the operator which shall satisfy the prechecks of the start up sequence.

The automatic sequence control shall be divided in at least seven (7) major successive sequences whose each intermediate state corresponds to a basic state of the plant:

- Auxiliary cooling system in service
- Condensate and feed water plant in service
- Boiler water system in service
- Gas Turbine start up and shut down
- Steam turbine start up and shut down
- Generator excitation
- Steam turbine load operation
- Automatic changeover to gas with bumpless or trip

Prior to the HRSG start up sequence, the water levels in the HP and LP drums will be established, the duty boiler circulation pumps (if used) and duty feed pumps will be running.

Prior to the turbine automatic start up sequence taking over control, the turbine shall be on barring, the AC auxiliary oil pump will be running, gland sealing and minimum start-up condenser back pressure will have been established and the main auxiliary cooling systems will be in operation. Providing initial prechecks are satisfied then the run up sequence shall proceed until the demand load is achieved.

Each of the seven (7) major sequences may be selected by the operator from the plant coordination control.

Reverse shut down sequence may also be initiated by the operator or by major fault. In the later case, the automatic system shall act in parallel with and as a complement to the protection system and allow the plant to be brought back to one of the seven (7) basic states mentioned above.

5. **HRSG CONTROL**

The main control functions to be implemented within the HRSG functional groups are LP feed control (and forced circulation if included), HP feed control (and forced circulation if included), and start up/load rate supervision and control. The latter regulates the turbine exhaust gas (TEG) flow through the boiler to limit the thermal stress within the critical boiler components during transient operating conditions.

The start up, operation within the normal load range and shut down of the HRSG shall be fully automated up to the functional group level. However, the initial HRSG filling operation, the establishment of initial HRSG drum level and forced circulation (if included) shall be manually controlled and supervised from the Operator station. In addition, only local control may be provided for minor drain and vent valves, where these are not automatic and which are not required during normal steam raising.

The HRSG shall be capable of constant pressure or variable pressure operation throughout its load range.

The incidence of a steam turbine trip (at any load), excessive rate of increase of the HP superheater temperature within the HRSG or a HRSG trip due to say a loss of feed water supply shall initiate the appropriate interlocks to move the dampers to the bypass position and not trip the gas turbine unless the dampers fail to move to the correct position within a preset time.

HRSG permissive signals will be required for opening the dampers from the bypass position.

The following shall move the dampers to the bypass position

- (i) All Feed Water Pumps Trip
- (ii) Low-Low Drum Level
- (iii) HRSG Exhaust Duct Pressure High Trip

The Contractor shall provide such additional control functions and interlocks as he deems necessary for the safe operation of the plant.

5.1 TURBINE EXHAUST GAS CONTROL (TEG)

All control and instrumentation systems required for the regulation and supervision of the heat input to the HRSG shall be associated with this control group. These shall include the controls of diverter damper position for the direct regulation of Turbine Exhaust Gas (TEG) flow through the boiler as well as all temperature and pressure measurement both on the gas and feed/steam sides considered necessary for the supervision of the HRSG.

The control system shall regulate the TEG flow into the HRSG to achieve the maximum rate of initial steam raising steam load and delivery temperature variation compatible with the design thermal stress limitations associated with the critical HRSG and steam turbine components. Automatic HRSG start up shall be possible under the full spectrum of operating conditions which shall include the following

- (i) The start up to the HRSG from any initial temperature condition (i.e. cold to full load temperature) with the associated gas turbine generator operating under part or full load conditions. In addition to the start up requirements, the control system shall meet the disturbance conditions.
- (ii) It shall maintain maximum steam generation compatible with the turbine bypass capacity in the event of the steam turbine tripping to permit the reloading of the latter with the minimum of delay.
- (iii) It shall minimize the effect of a partial or total plant load rejection and maintain the boiler in a state which will minimize the delay in the subsequent re-connection and loading of the steam turbine.

The Contractor shall clearly describe his control philosophy to meet the above requirements and provide sufficient details of his control system to show how the control functions are implemented. Contractor shall identify the critical control parameters and indicate between which limits they must be maintained.

The control and supervisory functions of the following plant items are to be associated under this group.

- (i) Diverter damper
- (ii) Gas passages in HRSG and Main Stack
- (iii) HRSG LP economizer, drum and evaporator sections.
- (iv) HRSG HP economizer, drum, evaporator and superheater sections.

It shall be possible to initiate the HRSG start up and shut down functions automatically from the plant co-ordination control level and manually from the functional group control level.

Start up shall only proceed provided that an appropriate plant state has been established prior to initiation. Pre-checks shall include the position of main steam stop and drain valves, drum levels, circulation pump operation and TEG temperature. The control system shall generate a TEG demand. A HRSG pressure control acting directly on the TEG demand signal shall be provided to limit the HRSG pressure to a preset limit during steam raising until the pressure control function of either the turbine bypass or the turbine itself is fully operational. This facility may be covered by the start-up control function mentioned above.

5.2 LP & HP FEED WATER CONTROL

All control and instrumentation systems required for the regulation of the low pressure feed water supply to the HRSG shall be associated with this control group. These shall include the feed regulation valves, the drum instrumentation and the forced circulation pumps.

Three element control shall apply for control within the normal load range acting on feed regulation valves nominally rated for 25% to 100% MCR feed flow. For HRSG filling and initial steam rising, a “three element control” shall be provided for HRSG drum level regulation according to ANSI/ISA-S77.42. The system shall be designed to maintain the drum level within acceptable

limits under all anticipated HRSG load changes and disturbances such as transfer from the duty feed pump to the standby feed pump.

For initial start up and HRSG loads below 30% MCR the measured drum level shall be compared with the desired drum level and the resulting error shall be used to control the start up valve position.

The HRSG shall be protected against loss of feed water. Under conditions of low-low drum level the circulation pumps (if used) shall trip, the drum blowdown shall be shut off and the boiler stop sequence initiated to insolate it from the TEG flow. The 'low-low' LP drum level condition shall trip the duty of HP feed and inhibit the start of the standby.

Each pump shall be equipped with individual flow measuring devices and in the event that the flow through a running pump falls below a preset value a leak-off valve shall open to recirculate feed water back to the deaerator storage vessel.

Supervisory equipment shall be provided for the feed pump which shall include
but not be limited to the following

- Suction pressure*
- Suction temperature
- Bearing temperatures (pump and motor)
- Lube oil pressures*
- Lube oil temperatures
- Vibration
- Thrust bearing wear*
- Motor temperatures

All parameters shall be alarmed and those marked with an asterisk (*) shall automatically trip the feed pump.

6. **STEAM TURBINE CONTROL**

The control system associated with the steam turbine generator shall include the governor, supervisory, tripping and auto run-up control systems. All

necessary equipment shall be provided for the automatic run up, synchronizing and loading of the steam turbine generator in conjunction with the steam bypass system. As part of the start sequence controls a load rate control shall be provided to control the turbine loading until all the surplus HP and LP steam has been diverted from the bypasses, subsequent to which any necessary load limiting functions shall be provided if these are required for the protection of turbine generator.

6.1 STEAM BYPASS CONTROL

6.1.1 BYPASS PRESSURE CONTROL

The arrangement of the bypass is indicated on the feed and steam system specification drawing. The pressure set point controller shall be generated at the plant co-ordination level and shall follow the variable pressure characteristic.

6.1.2 BYPASS TEMPERATURE CONTROL

Individual attemperating spray control loops shall be provided for each bypass valve to desuperheat the down stream steam conditions.

Spray water shall also be directed to the point of steam dumping within the condenser shell to prevent high velocity steam impingement on any component and to suppress the extent of noise generated.

6.1.3 HP STEAM PRESSURE LIMITING CONTROL

In the normal operating mode with the HP steam pressure control exercised by the turbine governor valves, the HP bypass shall perform a pressure limiting function to come into operation in the event of a plant disturbance such as a steam turbine trip or load rejection. This function shall be achieved by tracking the set point to the associated master pressure controller by an agreed margin above the block sliding pressure characteristic.

6.1.4 HP BYPASS PROTECTION

The bypass and condenser shall be protected by the rapid closure of the associated up-stream isolation valves under the following fault conditions.

- (i) Condenser pressure and/or temperature high
- (ii) CW supply failure
- (iii) Spray water supply to steam bypass pressure reducing station failure
- (iv) Spray water supply to condenser steam dump station failure
- (v) Condensate extraction pump failure

6.2 STEAM PRESSURE CONTROL

Pressure control of the HP steam manifold shall be provided by regulation of the HP turbine governor valves. The necessary pressure control loops shall be implemented at the turbine group control level with the variable pressure set point generated within the plant co-ordination control level.

For HP pressure control the pressure control loop shall act directly on the narrow range governor speed set point (speeder gear) input. The time response of the resetting action of the pressure controller shall be adjustable so that the balance between short term frequency support and boiler pressure variation may be optimized.

The LP pressure controller shall act directly to regulate the interceptor valve position.

6.3 TURBINE OIL CONTROL

This control shall combine all the regulatory, sequence control and monitoring functions associated with the turbine lubrication oil system, governor relay oil system and the rotor barring gear. The group control sequence shall co-ordinate the operation of the AC jacking oil pumps, the oil coolers and the barring gear to bring the turbine to run-up initiation.

The control associated with each pump, each cooler and the barring gear shall be configured as separate sub-groups. The automatic standby functions particularly for the DC Emergency rundown and jacking oil pumps shall be implemented at the sub-group level so that no fault at the plant co-ordination control level can inhibit their operation. Individual controls and state indication for each pump shall be provided at the sub-group level on the Operator Station.

The turbine barring gear sub-group provides the remote start, stop and monitoring functions and depending on the design of the clutch mechanism may also include the automatic coupling and release functions.

6.4 VACUUM AND GLAND SEAL CONTROL

This control combines all regulating, sequence control and monitoring functions associated with the main condenser vacuum system. It co-ordinates the operation of the air ejectors, the gland sealing system and the vacuum breaking valves. The Bidder shall describe the sub-group structures and the degree of co-ordination proposed to meet the functional group control requirements.

The control functions for gland seal steam shall include:

- (i) Steam pressure and temperature control during start-up and normal operation
- (ii) Automatic transfer between steam sources (if this is required) at different load levels
- (iii) Gland steam temperature monitoring and protection
- (iv) Gland steam leakage flow monitoring for indication of gland condition and efficiency calculation purposes.

The state of the gland steam system shall be monitored via the Operator Station.

6.5 CONDENSATE CONTROL

This control shall co-ordinate the control of the condensate make up valve to the condenser and the condensate return valve to the reserve feed water tank to maintain the condenser level.

It shall also control the condensate recirculation control valve to ensure a minimum condensate flow through the extraction pumps, air ejector and gland steam condensers.

6.6 DEAERATOR CONTROL

Deaerator heating shall be achieved by the regulation of the steam supply from the LP steam manifold. Adjustable temperature or pressure set points shall be provided.

Deaerator level control shall be provided by the regulation of the condensate inlet flow valve position, sequence control and Protection Functions.

6.7 TURBINE AUTO RUN-UP AND LOADING CONTROL

The system shall be designed to ensure consistent turbine run-up control from barring to full load at maximum rates compatible with the thermal state on the turbine, the steam conditions applied and the allowable capacity of turbine bypasses.

The system shall include:

- (i) All necessary pre-checks to ensure that the minimum conditions for auto run-up initiations are satisfied.
- (ii) All necessary computational and supervisory monitoring routines for the regulation of run-up
- (iii) Automatic synchronization of the turbine-generator and plant loading control.
- (iv) Turbine load limiting during initial plant start up and if considered necessary during on load operation.

Run-up shall be capable of being initiated both by the plant co- ordination control level and by operator action from the Operator Station. Pre-checks shall be incorporated in the system to ensure that control is only accepted if the plant conditions are appropriate for a turbine run-up. Any pre-check discrepancies shall be indicated to the operator independent of whether manual or auto control is selected.

The Contractor shall list all pre-check conditions to be monitored by the system and describe the provisions made for their revision and extension if design changes are necessary.

On acceptance of control the 'Turbine Run-up System' shall regulate the rate of run-up to maintain the thermal stress in critical components, such as the

turbine rotor, within acceptable limits. The system shall be capable of modifying the run-up rate if significant changes arise in the critical parameters, such as HP steam temperatures, used to infer thermal stress or in the direct computation of the acceptable run-up rate whilst turbine run-up is in progress.

Similarly the system shall monitor the supervisory measurements and take appropriate corrective action by imposing speed holds or speed run backs in the event of unacceptable conditions such as high bearing vibrations or high condenser back pressure being detected. The system shall recognize the critical speed bands of the turbine-generator and ensure minimum transition rates are maintained within these bands.

During run-up, speed control shall be exercised by applying a suitable speed reference signal to the wide range speed governor.

The auto synchronizing system shall be provided as a separate sub-system whose function shall be capable of being coordinated by the turbine auto run-up system. A separate manual system shall be provided for manual synchronizing at the CCR.

The automatic synchronizing equipment shall make the final adjustments to the generator speed and voltage conditions to bring them within predetermined limits for synchronizing. The synchronizer shall initiate closure of the main breaker making due allowance, for the circuit breaker closing time.

The system shall compute and impose, at the time of synchronizing, the Plant load appropriate to the thermal state of the turbine and thereafter generate a load limit and/or a loading rate limit which if not exceeded, will maintain the critical turbine components within the design thermal stress limits.

The magnitude of the plant loads and subsequent loading rates shall take into account the turbine state, the management of the turbine bypass system and if applicable the transfer of control from the start up valves to the on-load governing valves.

The turbine load limiting function shall apply whilst the system is performing the pressure governing function. The Bidder shall state what provisions are made to ensure that loading limitations are not exceeded whilst the turbine is in the pressure governing mode.

Because the load limiting function and the run-up control function are very similar they may be combined in the same hardware system.

6.8 STEAM TURBINE CONTROL SYSTEM

The Bidder shall offer his standard equipment which has been proven over a number of years in equivalent service.

- (i) The steam turbine control system shall be of electro-hydraulic type. A design relaying on electrical signal linkage rather than mechanical-hydraulic linkage between the various governing valve groups is preferred.
- (ii) The integrity of the governor speed control loops shall be assured against component failure by a high level of circuit redundancy.

Designs offering a three channel electronic system controlling on the median channel, dual channel electronic system with auto backed up or a single or dual channel systems backed up by a comprehensive hydraulic governor are all acceptable alternatives. There shall not be unwanted interaction between the action of electronic and back up governors under transient conditions. The structure of the governor system shall be clearly described by the Bidder.

- (iii) The system shall allow for the complete failure of one main channel without causing an immediate shut-down or unsafe operation of the turbine.
- (iv) Each electronic control channel shall be built-up of a number of standard modules which shall be fully interchangeable between similar channels. The equipment associated with each channel shall be mutually electrically and physically segregated to minimize fault propagation and to facilitate on-line testing and maintenance of each channel. Any limitation of on-line testing and rectification shall be stated in the Bid Proposal.
- (v) The system hardware shall have well developed self-diagnostic fault detection features and clear fault finding procedures shall be provided to permit rapid on-line fault rectification by module replacement.

- (vi) The turbine speed feed back signals shall be derived from a toothed wheel and probe system. The toothed wheel shall be integrated with the turbine shaft. At least one separate speed probe shall be provided for each electronic control channel. All additional speed measurement channels shall be independent of the turbine control system and separate speed probes shall be provided. The probes shall be on-line replaceable without re-calibration being necessary.
- (vii) Facilities shall be included for the on-load testing of the governor valves which shall interface with the 'On-Load' valve test facilities as described herein.

An on-load test facility shall be provided for the following valves:

- The HP Main Stop Valves
- The HP Governor/Control Valves
- The LP Stop Valves
- The LP Control Valve

The tests shall be conducted from the Operator station where all interlocks to ensure that the steam turbine load has been suitably reduced for the tests and that only one steam path can be interrupted at any one time are indicated. A multi channel trend monitor shall be provided for recording both the valve position demand signal and the valve position feed back signal over the full range of travel, with sufficient resolution to display any significant non linear stroke characteristics.

The stop valves and control valves shall be tested in conjunction with the bypass system to ensure a steam path is maintained.

6.9 TURBINE TRIPPING SYSTEM

This system shall be provided to trip the turbine-generator under hazardous operating conditions. Its main function shall be to check the validity of the trip demand signals and to ensure trip action results in immediate response to a valid trip demand.

If electronic over-speed governing is offered then the over-speed governors shall either form part of this sub-group or from a separate sub-group. In this event, a three channel system shall be provided with a two out of three voting

trip action unless otherwise approved by the Owner.

The CPU of turbine protection shall be triple. Two of three conditions shall be adopted for trip turbine. No single CPU fail will trip turbine

6.10 TURBINE SUPERVISORY SYSTEM

Turbine supervision system shall be provided for the safe and efficient operational of the turbine generator units.

In general all associated indications required in the Operator Station shall be provided by direct reading instrumentation and records as well as VDU displays.

The supervision system shall include monitoring of the following items

- Turbine speed
- Shaft eccentricity
- Shaft vibration
- Differential expansion between casing and rotor
- Casing and rotor expansion
- Thrust bearing wear
- Valves position
- Turbine temperature monitoring

The shaft vibration system shall be part of a complete monitoring and analysis package.

7. INSTRUMENT DETAIL

The contractor shall provide details of the instrumentation and control for each Gas Turbine, HRSG and Steam Turbine and their associated auxiliaries.

8. TESTING

Full functional and performance tests shall be carried out in conjunction with the associated mechanical equipment.

During commissioning of the equipment, the Contractor shall run a test to ascertain that the equipment is performing properly in every respect.

Any deficiencies in the Contractor's equipment brought out during the test shall be promptly remedied by the Contractor at no expense to the Owner.

Any modifications, changes, adjustments or corrections, which may be necessary in the opinion of the Owner or as may be requested by the Owner, shall be done by the Contractor. This corrective work shall relate only to corrections of defects based upon the original design of the equipment and shall be done without additional cost to the Owner.

9. **QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)**

The Contractor shall conform to the quality assurance/quality control requirement of Part 4, Subsection 2.7 Clause 4 unless the mechanical equipment with which the Instrumentation and Control are associated is Level 1 in which case Clause 3 shall be applicable.

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